

1 (a)(i)

$$\begin{aligned}s &= \frac{1}{2} (u + v) t \\&= \frac{1}{2} (4.0 + 37.0)(17.0) \\&= 348.51 \\&\approx 349 \text{ m}\end{aligned}$$

[C1]
[A1]

1 (a)(ii)

$$\begin{aligned}\Delta t &= 0.6 / 100 \times 17 = 0.1 \text{ s} \\s_{\max} &= \frac{1}{2} (4.1 + 37.1)(17.1) = 352.26 \\s_{\min} &= \frac{1}{2} (3.9 + 36.9)(16.9) = 344.76 \\2\Delta s &= s_{\max} - s_{\min} = 352.26 - 344.76 = 7.4 \\ \Delta s &= 3.7 \approx 4\end{aligned}$$

[C1]

$$\frac{\Delta s}{s} \times 100\% = \frac{4}{348.51} \times 100\% = 1.1\%$$

[A1]

Alternatively

$$\begin{aligned}s &= \frac{1}{2} (u + v) t \\ \frac{\Delta s}{s} &= \frac{\Delta(u + v)}{(u + v)} + \frac{\Delta t}{t}\end{aligned}$$

[C1]

$$\frac{\Delta s}{s} \times 100\% = \left(\frac{0.1 + 0.1}{4.0 + 36.0} \right) \times 100\% + 0.6\%$$

$$\frac{\Delta s}{s} \times 100\% = 1.1\%$$

[A1]

1 (a)(iii)

$$\frac{\Delta s}{s} \times 100\% = 1.1\%$$

$$\Delta s = 0.011 \times 349$$

$$\Delta s = 3.84 \approx 4 \text{ m}$$

$$s = (349 \pm 4) \text{ m}$$

[B1]

1 (b)(i)

$$\begin{aligned}\Delta KE &= \frac{1}{2} mv^2 - \frac{1}{2} mu^2 \\&= \frac{1}{2} \times 95 \times [(37.0)^2 - (4.0)^2] \\&= 64300 \text{ J}\end{aligned}$$

[C1]
[A1]

1 (b)(ii)

$$\begin{aligned}\Delta GPE &= \Delta mgh \\&= 95 \times 9.81 \times -349 \sin 40^\circ \\&= -209000 \text{ J}\end{aligned}$$

[C1]
[A1]

1 (b)(iii)

$$\begin{aligned}GPE_i + KE_i + W_{\text{friction}} &= GPE_f + KE_f \\ \text{Work done by friction} &= \Delta GPE - \Delta KE \\&= -209100 + 64300 \text{ J} \\&= -144800\end{aligned}$$

[C1]

$$f \times 349 \cos 180^\circ = -144800$$

$$f = 415 \text{ N}$$

[A1]

2 (a)